Qwen Model Fine-Tuning Guide

https://gemini.google.com/app/99ada6e2c65fc7b4

User prompt: about this fine tuning example is there suggested qwen model for me to work with?

Response: Yes, you can readily adapt this Hugging Face Agents Course fine-tuning notebook to use a Qwen model.

Suggested Qwen Models

- **`Qwen/Qwen2.5-3B-Instruct`**: A great balance for the free or standard Colab tier, fitting nicely within memory constraints while providing strong baseline instruction-following and tool-calling capabilities.
- **`Qwen/Qwen2.5-7B-Instruct`**: Ideal if you have access to a slightly larger GPU (such as Colab Pro or an A100) for enhanced reasoning performance.
- **`Qwen/Qwen2.5-1.5B-Instruct`**: A lightweight alternative if you want faster training times and lower VRAM usage on resource-constrained hardware.

Key Code Modifications Needed

If you switch to a Qwen model, you will need to update a few lines in the notebook:

1. **Model Name Update (Step 5):** Change the model reference to your chosen Qwen checkpoint:

   ```
   model_name = "Qwen/Qwen2.5-3B-Instruct"
   ```

2. **Chat Template Handling (Steps 5 & 8):** The notebook overrides the tokenizer chat template specifically for Gemma's `<start_of_turn>` tags. For Qwen, you should adjust the `tokenizer.chat_template` or preprocessor logic to align with Qwen's chat template format while preserving the custom `<think>` and `<tool_call>` XML tags introduced in the dataset.
3. **LoRA Target Modules (Step 9):** Qwen architectures use standard attention and MLP projection layers (`q_proj`, `k_proj`, `v_proj`, `o_proj`, `gate_proj`, `up_proj`, `down_proj`), so the existing LoRA target modules configuration will map over cleanly.

Fine-Tuning an LLM for Function-Calling Walkthrough

This video provides a helpful visual walkthrough of the exact Hugging Face Agents Course notebook you are referencing.
